SALMAN SHAFI

Ottawa, ON | salshafi44@gmail.com | github.com/salshafi | linkedin.com/in/salmanshafi44

SUMMARY

Seeking a 4-month co-op, January–April 2027 (available on site in Ottawa). Software Engineering Technology (AI) co-op student in Semester 3 who works with SQL and SQLite databases and Python data analysis (pandas, scikit-learn), builds REST APIs in Python and C#, and automates tasks on Linux with Bash. Self-taught in SQL and databases, with a long-term goal of AI/ML engineering.

TECHNICAL SKILLS

- **Languages:** SQL, Python, Java, C#, JavaScript (ES6), HTML/CSS
- **Data & ML:** pandas, scikit-learn, matplotlib (linear regression, train/test split, R-squared, data cleaning and visualization)
- **Backend:** SQLite, SQLAlchemy, Entity Framework Core, JDBC, FastAPI, Pydantic, ASP.NET Core Web API, REST API design, pytest
- **Linux & Automation:** Ubuntu (VM), Bash scripting, cron, grep (regex), sed, awk, find, Apache/MySQL via XAMPP
- **Tools:** Git/GitHub, VS Code, Eclipse, React (Vite), Gemini API
- **Currently learning:** Node.js/Express, MongoDB, JWT authentication, Docker, cloud fundamentals

PROJECTS

job-tracker-api | Python, FastAPI, SQLAlchemy, SQLite, pytest — github.com/salshafi/job-tracker-api

- Built a REST API that stores and queries application data in a SQLite database through SQLAlchemy, with full CRUD, status filtering and partial updates (PATCH), and Pydantic validation that returns 422 on bad input.
- Added an AI assistant endpoint that uses the Gemini API to answer questions about a user's applications, with the API key kept server-side.
- Wrote 27 pytest tests against an in-memory database, plus CORS support, interactive API docs and a README.

Java Transaction Engine | Java SE, JDBC, SQLite, ExecutorService — github.com/salshafi/java-transaction-engine

- Built a console app that processes a batch of financial transactions concurrently on a fixed pool of 3 worker threads.
- Saved transactions to SQLite through JDBC using parameterized PreparedStatement queries.
- Applied OOP design: an interface with an implementing class and a custom InsufficientFundsException that rejects over-limit transactions.

GradeTracker | C# Web API, EF Core, SQLite, JavaScript — github.com/salshafi

- Built a C# Web API with Entity Framework Core and SQLite, plus a JavaScript front end.

Linear Regression Data Analysis | Python, pandas, scikit-learn, matplotlib — COMP237

- Cleaned, explored and visualized two datasets (2,362 e-commerce transactions; fish measurements) with histograms, scatterplots and boxplots.
- Engineered features, one-hot encoded categories, split data 70/30 and trained linear regression models, evaluating them with R-squared.

Linux Text Processing & Cron Automation | Bash, grep, sed, awk, find, cron — COMP301

- Used grep (basic and extended regex), sed (in-place edits with backups) and awk (field filtering, counting).
- Scheduled a recurring job with cron using a safe crontab edit workflow; fixed Windows CRLF line-ending errors with sed.

EDUCATION

Software Engineering Technology – Artificial Intelligence (Co-op) — Winter 2026 – Present

Centennial College, Toronto, ON (online) | Currently in Semester 3

- **Current courses (Fall 2026):** Web Application Development (COMP229, MERN), Java Programming (COMP228), Introduction to AI (COMP237), Linear Algebra & Statistics
- **Completed coursework:** Python, Linux Shell Scripting (COMP301), C# OOP (COMP123), Client-Side JavaScript (COMP125)

EXPERIENCE

Customer Service Associate (Self-Checkout) – Independent Grocery Store — 2018 – 2021

- Troubleshot scanning and payment errors across multiple self-checkout stations at once.
- Verified age-restricted purchases, handled voids and overrides, and de-escalated frustrated customers.